Thermal Feedback in a Microscopic Black Hole Reactor

Stability of a coupled Hawking–Bondi model

Emir Anaraly uulu
Independent Researcher, Bishkek, Kyrgyz Republic

Continuation of the fixed-medium study | Revised: 30 September 2026

Abstract

The fixed-medium Hawking–Bondi mass balance studied in Paper I is unstable. Does heating of the accreting medium change that result? We couple mass to an effective environmental temperature: ideal-gas sound speed makes the Bondi coefficient proportional to $T^{-3/2}$, while deposited Hawking power and linear cooling determine the temperature. The resulting two-dimensional system has exactly one positive equilibrium. Heating raises its temperature and mass, but its Jacobian determinant is negative for every positive feedback and cooling strength; the equilibrium is a saddle. More generally, the same conclusion holds for any one-zone model in which heating decreases with mass, accretion decreases with temperature, and cooling increases with temperature. We distinguish the destabilizing thermal coupling at a fixed equilibrium from the slower departure that can accompany a shifted equilibrium. Phase-space trajectories and parameter scans illustrate the analytic results. This is a theorem about the stated reduced model, not a prediction for a real surrounding material or a hydrodynamic accretion flow.

1 Introduction and scope

In the first stage of this project, we balanced an idealized Bondi accretion rate against a simplified Hawking evaporation rate while holding the surrounding medium fixed [5]. The mass equation has one positive balance, but its right-hand side increases with mass: either sign of a small perturbation drives the system away from that balance. Holding the medium fixed is particularly restrictive when the hole is itself a strong source of radiation.

Here we ask whether Hawking heating can turn the balance into an attracting state. We make the smallest extension that allows this question to be tested: mass $M(t)$ and one effective environmental temperature $T(t)$ evolve together, while density remains prescribed. The temperature changes the sound speed in the Bondi coefficient. This yields a tractable stability calculation, a phase portrait, and a direct comparison with the fixed-medium result.

Recent time-dependent calculations of primordial-black-hole accretion in stellar interiors show how radiative cooling can modify an adiabatic Bondi flow [6]. Their stellar gas, accretion-powered radiation, and spatial hydrodynamics differ from the Hawking-heated reservoir considered here. We use that work as motivation to let thermodynamics affect accretion, without importing its rates or thresholds.

We retain an uncharged, nonrotating black hole and the photon-blackbody evaporation baseline of Paper I. The model does not describe confinement, power collection, material survival, or a realizable reactor. Throughout, "stability" refers to the mass–temperature equilibrium of the specified differential equations.

2 From the fixed medium to a thermal reservoir

2.1 Mass gain and loss

For a Schwarzschild black hole, Hawking's temperature scales as $T_H \propto M^{-1}$ [1]. As in Paper I, approximating its photon power by blackbody emission from the horizon area gives

$$P_{\rm bb} = \frac{K}{M^2}, \qquad K = \frac{\hbar c^6}{15360\pi G^2}, \qquad \dot{M}_{\rm evap} = -\frac{B}{M^2}, \quad B = \frac{K}{c^2}.$$

This power law is a deliberately simplified baseline. Greybody factors and available particle species alter realistic emission rates [2, 3]. For steady, spherical Bondi accretion of a gas at rest far from the hole [4],

$$\dot{M}_{\rm acc} = A(T)M^2, \qquad A(T) = \frac{4\pi\lambda\rho_\infty G^2}{c_s(T)^3}.$$

At fixed density ρ_∞, fixed accretion parameter λ, and constant ideal-gas composition and adiabatic index,

$$c_s^2 = \frac{\gamma k_B T}{\mu m_p}, \qquad A(T) = A_0\left(\frac{T}{T_0}\right)^{-3/2}.$$

The effective T must represent the gas relevant to the Bondi boundary condition. Heating only a tiny region next to the horizon would not, by itself, justify changing the distant sound speed. We assume a well-mixed reservoir that couples the deposited radiation to that gas; this is a closure assumption, not a radiative-transfer result.

Paper I used $\rho_\infty = 10^3\,\mathrm{kg\,m^{-3}}$, $c_s = 1500\,\mathrm{m\,s^{-1}}$, and $\lambda = 1/4$ as a formal water-like reference. Those numbers cannot assign an ideal-gas T_0 consistently. We use its characteristic mass and time only as mathematical scales and do not calibrate the new thermal parameters to water.

2.2 Deposited power and cooling

Let a fraction ϵ of the baseline luminosity enter an effective reservoir with constant heat capacity $C_{\rm env}$. A linear cooling law returns it toward T_0 on a timescale τ_c. The dimensional model is

$$\boxed{\begin{aligned} \dot{M} &= A_0\left(\frac{T}{T_0}\right)^{-3/2} M^2 - \frac{B}{M^2}, \\ C_{\rm env}\dot{T} &= \frac{\epsilon K}{M^2} - \frac{C_{\rm env}}{\tau_c}(T - T_0). \end{aligned}} \tag{1}$$

All constants are positive, $0 < \epsilon \le 1$, and $M, T > 0$. The deposited fraction is a model parameter; it has not been derived from the high-energy spectrum, opacity, or geometry. The reservoir also suppresses spatial structure and the energy carried by gas entering and leaving it.

2.3 Dimensionless control parameters

Let $M_{\rm c0} = (B/A_0)^{1/4}$ be the balance mass of the fixed-medium equation and $t_* = (A_0 M_{\rm c0})^{-1}$ its time unit. With

$$x = \frac{M}{M_{\rm c0}}, \qquad \theta = \frac{T}{T_0}, \qquad s = \frac{t}{t_*},$$

Eq. (1) becomes

$$\boxed{\begin{aligned} \frac{dx}{ds} &= F(x,\theta) = \theta^{-3/2}x^2 - x^{-2}, \\ \frac{d\theta}{ds} &= G(x,\theta) = \eta x^{-2} - \zeta(\theta - 1), \end{aligned}} \tag{2}$$

where

$$\eta = \frac{t_*\epsilon K}{C_{\rm env}T_0 M_{\rm c0}^2}, \qquad \zeta = \frac{t_*}{\tau_c}, \qquad \chi = \frac{\eta}{\zeta} = \frac{\epsilon P_{\rm bb}(M_{\rm c0})\tau_c}{C_{\rm env}T_0}.$$

The ratio χ measures deposited energy during one cooling time against the reservoir's thermal energy scale. The rate ζ measures cooling relative to the fixed-medium mass time unit. In a two-parameter scan, χ and ζ can vary independently in the equations; changing τ_c alone would change both.

3 Where the balance moves

The two nullclines of Eq. (2) are

$$F = 0: \quad x = \theta^{3/8}, \qquad G = 0: \quad \theta = 1 + \chi x^{-2}.$$

The first increases with temperature and the second decreases with mass. Their intersection (x_*, θ_*) satisfies

$$\boxed{x_* = \theta_*^{3/8}, \qquad \theta_* - 1 = \chi \theta_*^{-3/4}.} \tag{3}$$

The function $h(\theta) = \theta - 1 - \chi\theta^{-3/4}$ is negative at 1, tends to infinity, and has $h'(\theta) = 1 + (3\chi/4)\theta^{-7/4} > 0$. Thus exactly one positive equilibrium exists, with $\theta_* > 1$ and $x_* > 1$, for every $\chi > 0$. Its location depends on χ and not separately on ζ.

The limiting forms also follow from Eq. (3):

$$\begin{aligned} &\chi \ll 1: \quad \theta_* = 1 + \chi + O(\chi^2), \quad x_* = 1 + \tfrac{3}{8}\chi + O(\chi^2), \\ &\chi \gg 1: \quad \theta_* \sim \chi^{4/7}, \quad x_* \sim \chi^{3/14}. \end{aligned}$$

Heating therefore moves the equality of mass gain and loss. This geometric shift does not determine whether trajectories return to it.

4 Stability of the coupled equilibrium

4.1 A saddle for every positive coupling

Linearizing Eq. (2) about (x_*, θ_*) gives $d(\delta x, \delta\theta)^{\mathsf{T}}/ds = J_*(\delta x, \delta\theta)^{\mathsf{T}}$. Since $\theta_*^{-3/2} = x_*^{-4}$,

$$J_* = \begin{pmatrix} a & b \\ c & -\zeta \end{pmatrix}, \qquad a = 4x_*^{-3} > 0, \quad b = -\frac{3}{2}\theta_*^{-5/2}x_*^2 < 0, \quad c = -2\eta x_*^{-3} < 0.$$

Its determinant is

$$\boxed{\det J_* = -4\zeta x_*^{-3} - 3\eta\theta_*^{-5/2}x_*^{-1} < 0.} \tag{4}$$

The eigenvalues are real and have opposite signs. Hence the unique equilibrium is a saddle for every $\eta, \zeta > 0$; neither a stable node nor a Hopf bifurcation is possible within this model.

There is a useful distinction between two comparisons. At the *new* equilibrium, freeze θ at θ_* while perturbing the mass: the mass growth rate is $a > 0$. Once temperature can respond, $bc > 0$ and

$$\lambda_\pm = \frac{a - \zeta \pm \sqrt{(a+\zeta)^2 + 4bc}}{2}, \qquad \lambda_+ > a > 0, \quad \lambda_- < -\zeta. \tag{5}$$

Thermal coupling amplifies the unstable rate relative to freezing temperature *at the same equilibrium*. Comparing instead with the *old* fixed-medium equilibrium gives its rate 4. For $\chi = \zeta = 1$, the new equilibrium has $a = 2.2341$ and $\lambda_+ = 2.4315$: thermal coupling raises 2.2341 to 2.4315, while the shifted system still departs more slowly than the old rate 4. This distinction prevents a misleading claim that a smaller λ_+ means the feedback is stabilizing.

The unstable eigendirection has slope $(\lambda_+ - a)/b < 0$ in the (x, θ) plane. The stable one has slope $(\lambda_- - a)/b > 0$. Only initial conditions on the one-dimensional stable manifold approach the saddle; generic nearby perturbations have an unstable component. In particular, a downward mass perturbation at initially fixed temperature increases the Hawking heating term and weakens accretion, reinforcing mass loss.

For fixed $\zeta > 0$ and very large χ, Eq. (5) gives $\lambda_+ \sim 7\chi^{-9/14}$. It becomes small but stays positive at every finite χ. This asymptotic is a formal parameter limit, not a prediction that arbitrarily strong heating is physically achievable.

At fixed χ, changing ζ changes $\eta = \chi\zeta$ as well. The slow- and fast-response limits are

$$\lim_{\zeta\to 0^+} \lambda_+ = a, \qquad \lim_{\zeta\to\infty} \lambda_+ = a + 3\chi\theta_*^{-5/2}x_*^{-1} > a.$$

The fast limit follows by keeping $G = 0$ as the temperature responds to mass; the slow limit freezes its response over a mass-evolution time. Neither reverses the instability.

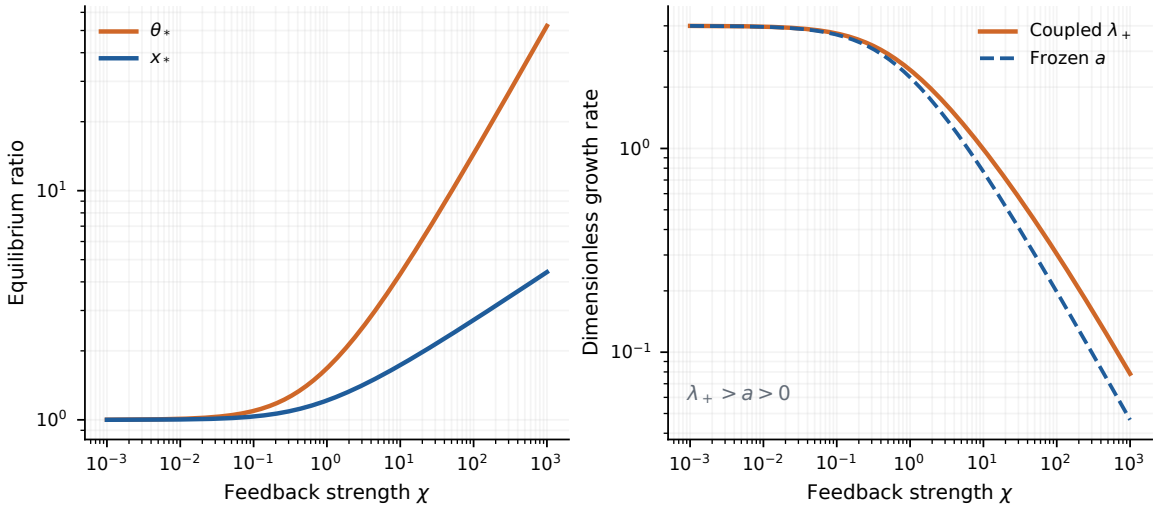


Figure 1: Heating shifts the balance without making it attracting. Left: equilibrium temperature and mass from Eq. (3). Right: for $\zeta = 1$, the positive eigenvalue λ_+ and the mass growth rate $a = 4x_*^{-3}$ with temperature held at θ_*. The thermal response increases the local growth rate relative to that frozen-temperature comparison, although both rates fall as the equilibrium moves to larger mass. All quantities are dimensionless.

4.2 A sign theorem beyond the chosen power laws

The saddle conclusion is not tied to linear cooling. Consider a differentiable one-zone system

$$\dot{M} = \mathcal{F}(M,T) = A(T)M^2 - BM^{-2}, \qquad \dot{T} = \mathcal{G}(M,T) = H(M) - \Lambda(T).$$

At any positive equilibrium, if $A > 0$, $A'(T) < 0$, $H'(M) < 0$, and $\Lambda'(T) > 0$, then

$$\mathcal{F}_M = 2A(T)M + 2BM^{-3} > 0, \quad \mathcal{F}_T = A'(T)M^2 < 0, \quad \mathcal{G}_M = H'(M) < 0, \quad \mathcal{G}_T = -\Lambda'(T) < 0.$$

Consequently $\det D(\mathcal{F},\mathcal{G}) = \mathcal{F}_M\mathcal{G}_T - \mathcal{F}_T\mathcal{G}_M < 0$. Every positive equilibrium in this class is a saddle. The theorem asserts local instability where an equilibrium exists; a more elaborate model may violate these sign assumptions or require additional state variables.

5 Numerical phase space and departure

5.1 Method and checks

We solve Eq. (3) by a bracketed root finder, evaluate J_* analytically, and integrate Eq. (2) with `scipy.integrate.solve_ivp` (DOP853, relative tolerance 10^{-10}, absolute tolerance 10^{-12}, maximum step 0.01). Trajectories stop before x approaches the singular boundary at zero or leaves the plotting window. The supplied script regenerates all figures, records the table below, and checks equilibrium residuals and eigenvalue signs. These numerical checks test the equations, not the physical validity of the closure.

For $\chi = \zeta = 1$, the root and linearized rates are

$$(x_*, \theta_*) = (1.214276, 1.678211), \qquad (\lambda_-, \lambda_+) = (-1.197336, 2.431462).$$

Figure 2 shows the two nullclines and trajectories near their crossing. We integrate short segments of the local stable manifold backward and of the unstable manifold forward from eigenvector displacements near the saddle. These curves illustrate the local geometry within the plotted domain; they do not establish a global basin boundary.

5.2 What an equal percentage perturbation compares

Figure 3 applies ±2% mass perturbations to each system's *own* equilibrium. The fixed-medium trajectories begin at $(x,\theta) = (1 \pm 0.02, 1)$; the coupled ones begin at $(x,\theta) = (x_*(1 \pm 0.02), \theta_*)$.

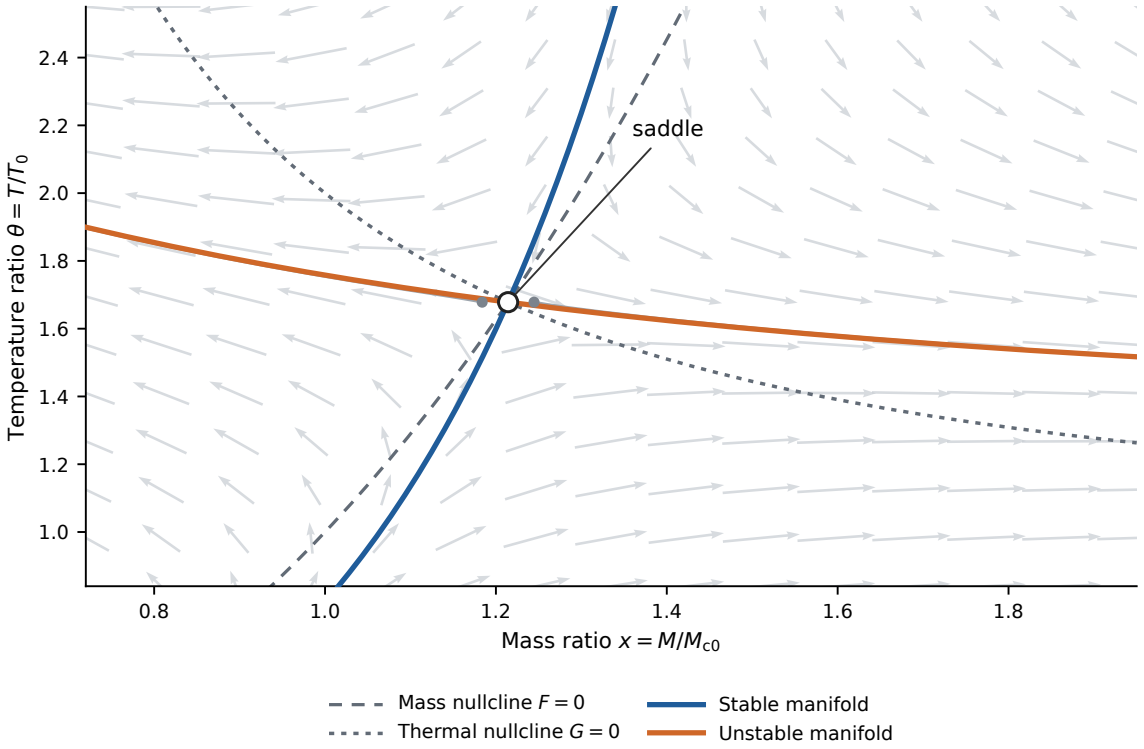


Figure 2: One crossing, two different directions. Phase portrait at $\chi = \zeta = 1$. Gray dashed lines are the mass and thermal nullclines; the circle marks their intersection. Blue curves approach the saddle forward in time along its stable manifold, whereas orange curves depart along its unstable manifold. Thin gray curves start from nearby mass perturbations at fixed equilibrium temperature. Axis ranges stop well before the singular mass boundary.

Table 1: Equilibria and linear rates for $\zeta = 1$. The rate a freezes temperature at each new equilibrium; λ_+ includes the temperature response. Values are dimensionless and rounded from the reproducibility script.

χ	θ_*	x_*	a	λ_-	λ_+
0.01	1.00993	1.00371	3.95580	−1.00588	3.96168
0.1	1.09352	1.03409	3.61728	−1.04971	3.66699
1	1.67821	1.21428	2.23413	−1.19734	2.43146
10	4.33092	1.73268	0.76897	−1.22271	0.99167
100	14.47513	2.72417	0.19786	−1.10595	0.30381

Both use $s = t/t_*$, and the vertical axis divides mass by the relevant equilibrium mass. The curves compare dimensionless departures, not two physical environments with the same initial black-hole mass. Changing χ changes the equilibrium itself, and no material calibration is implied.

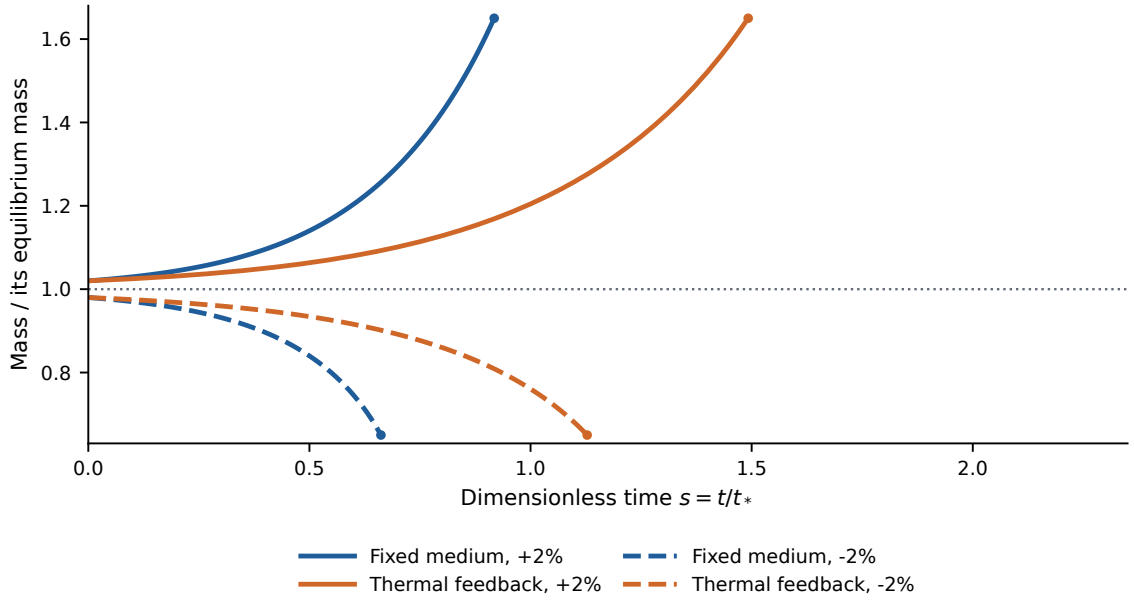


Figure 3: Perturbations leave both balances. Mass divided by each model's own equilibrium mass following ±2% perturbations. Solid curves start above equilibrium; dashed curves start below. The coupled model has $\chi = \zeta = 1$ and initially unperturbed equilibrium temperature. Endpoints mark plotting cutoffs, not complete evaporation or unbounded accretion.

The scan in Table 1 makes the distinction in Eq. (5) visible. For every listed χ, $\lambda_+ > a > 0$, although the absolute rate drops as the equilibrium shifts. A 41×41 grid with $10^{-2} \leq \chi, \zeta \leq 10^2$ has $\lambda_+ > 0$ at every sampled point; the analytic determinant, rather than this finite scan, establishes the sign for all positive parameters.

6 Physical meaning and limitations

The negative determinant has a simple feedback interpretation. A decrease in M raises the modeled Hawking power. Deposited heat raises T, which raises sound speed and decreases the Bondi coefficient at fixed density. Accretion then replaces less of the evaporated mass. The reverse perturbation tends toward greater accretion and weaker heating. Cooling slows and shapes this response but does not reverse its sign.

The conclusion is restricted by the closure assumptions. In particular, the Bondi coefficient uses properties of gas outside the accretion region; a single reservoir temperature cannot represent a radial radiation field, velocity profile, or changing density. A real material's deposition fraction and cooling depend on frequency, composition, optical depth, and geometry. The photon-blackbody approximation omits greybody factors and particle thresholds. The water-like reference in Paper I is not an ideal gas, and its smallest reference Bondi scales challenge continuum flow [5]. These issues prevent the dimensionless values in Table 1 from being interpreted as a reactor design or a lifetime prediction.

The radiative stellar-core calculations of Cantiello and collaborators solve a much richer accretion problem and report that cooling can enhance the accretion rate in part of their regime [6]. Their radiative output comes from the accreting gas, whereas our heating term is externally modeled Hawking power proportional to M^{-2}. Thus their quantitative results cannot be substituted for ϵ, $C_{\rm env}$, or τ_c here.

A next model could evolve density together with temperature and calculate where Hawking energy is deposited. The sign theorem identifies what that extension must check: whether the

effective gain, heating, and cooling derivatives still have the signs assumed above. It does not promise that an additional variable will create stability.

7 Conclusion

Paper I found a repelling Hawking–Bondi mass balance in a prescribed medium. Coupling that mass to a heated, cooling reservoir yields one hotter, larger balance. The exact determinant in Eq. (4) is negative for all positive feedback and cooling strengths, so this balance is a saddle. The same local result follows for a broader class of monotone one-zone heating and cooling laws.

The shifted equilibrium can have a smaller positive growth rate than the old fixed-medium balance, even though allowing temperature to respond makes it more unstable than freezing temperature at the new balance. Numerical trajectories exhibit both effects. Whether spatial transport, density evolution, and realistic emission change the behavior of a physical system remains a separate question.

A Scaling and the strong-feedback limit

At $M_{\rm c0}$, $B = A_0 M_{\rm c0}^4$ and $t_*^{-1} = A_0 M_{\rm c0}$. Substitution of $M = M_{\rm c0} x$, $T = T_0 \theta$, $t = t_* s$ into Eq. (1) gives Eq. (2). The source strength η uses $\epsilon K/(C_{\rm env} T_0 M_{\rm c0}^2)$, which has units of inverse time; $\chi = \eta/\zeta$ is dimensionless.

For $\chi \gg 1$, Eq. (3) implies $\theta_*^{7/4} \sim \chi$ and $x_* \sim \chi^{3/14}$. Hence $a = 4x_*^{-3} \sim 4\chi^{-9/14}$. The off-diagonal product is $bc = 3\eta\theta_*^{-5/2} x_*^{-1} \sim 3\zeta\chi^{-9/14}$. With fixed positive ζ, the positive root of Eq. (5) obeys $\lambda_+ \sim a + bc/\zeta \sim 7\chi^{-9/14}$. This limit is nonuniform if ζ is also taken to zero.

B Reproducibility

The included `reproduce.py` needs NumPy, SciPy, and Matplotlib. Run `python reproduce.py` in the `paper` directory, then `pdflatex main.tex` twice to regenerate the plots and manuscript. The script evaluates the roots, Jacobian, eigenvalues, and numerical trajectories, and writes `results.json`. Plotting terminates trajectories at explicit bounds away from $x = 0$. Numerical tolerances and stop conditions are specified in the script; no uncertainty estimate for a physical medium is inferred from solver precision.

References

[1] S. W. Hawking, "Particle creation by black holes," *Communications in Mathematical Physics* **43**, 199–220 (1975). doi:10.1007/BF02345020.

[2] D. N. Page, "Particle emission rates from a black hole: Massless particles from an uncharged, nonrotating hole," *Physical Review D* **13**, 198–206 (1976). doi:10.1103/PhysRevD.13.198.

[3] J. H. MacGibbon and B. R. Webber, "Quark- and gluon-jet emission from primordial black holes: The instantaneous spectra," *Physical Review D* **41**, 3052–3079 (1990). doi:10.1103/PhysRevD.41.3052.

[4] H. Bondi, "On spherically symmetrical accretion," *Monthly Notices of the Royal Astronomical Society* **112**, 195–204 (1952). doi:10.1093/mnras/112.2.195.

[5] E. Anaraly uulu, "Stability of a Microscopic Black Hole Reactor: Hawking evaporation and a fixed-medium Bondi model," independent manuscript (2026), https://github.com/IPhO-KG/microscopic-black-hole-reactor.

[6] M. Cantiello, O. Gottlieb, C. Norton, M. Kleban, and K. Van Tilburg, "Accretion of Primordial Black Holes in Stellar Interiors," arXiv:2606.02726 (2026), https://arxiv.org/abs/2606.02726.